

L6 Control Quiz ANS

1. What control commands does the controller output to follow the trajectory from motion planning?

- A) Brake, acceleration, and steering commands
- B) Only a map-level route
- C) Object-detection bounding boxes
- D) HD-map updates

ANS: A) Brake, acceleration, and steering commands

2. In feedback control, what is the tracking error $e(t)$?

- A) The sum of reference and output
- B) The reference signal minus the system output
- C) The derivative of the reference only
- D) The control input minus the system state

ANS: B) The reference signal minus the system output

3. Which PID term uses the history of the tracking error and helps eliminate steady-state error?

- A) Proportional term
- B) Derivative term
- C) Integral term
- D) Prediction term

ANS: C) Integral term

4. What is a typical effect of increasing the derivative gain K_d ?

- A) A large increase in steady-state error
- B) Elimination of all measurement noise
- C) Increased overshoot
- D) Reduced overshoot with greater sensitivity to measurement noise

ANS: D) Reduced overshoot with greater sensitivity to measurement noise

5. What does rise time measure in a step response?

- A) The time for the transient response to rise close to the steady-state value
- B) The time at which maximum overshoot occurs
- C) The time required to compute the control law
- D) The final difference between input and output

ANS: A) The time for the transient response to rise close to the steady-state value

6. What is maximum overshoot?

- A) The time after which the output remains in a specified band
- B) The amount by which the maximum transient response exceeds the steady-state value
- C) The difference between two controller gains
- D) The minimum value reached by the reference signal

ANS: B) The amount by which the maximum transient response exceeds the steady-state value

7. What is the main characteristic of the proportional PID term?

- A) It depends only on accumulated past error
- B) It predicts future states over a horizon
- C) Its contribution is proportional to the current tracking error
- D) It is independent of the tracking error

ANS: C) Its contribution is proportional to the current tracking error

8. According to the slides, increasing K_p generally produces which combination?

- A) Slower response and larger steady-state error
- B) No change in response
- C) Lower overshoot in all cases
- D) Faster response, potentially more overshoot and oscillation

ANS: D) Faster response, potentially more overshoot and oscillation

9. What alternative name is used for Model Predictive Control?

- A) Receding Horizon Control
- B) Open-Loop Integral Control
- C) Static Gain Scheduling
- D) Randomized Feedback Control

ANS: A) Receding Horizon Control

10. At each MPC time step, how much of the optimized control sequence is normally applied?

- A) The entire sequence without further optimization
- B) Only the first control command
- C) Only the last control command
- D) No control command until the horizon ends

ANS: B) Only the first control command

11. Why did MPC decelerate earlier than PID in the sharp-turn example?

- A) MPC ignored the reference trajectory
- B) MPC used no vehicle model
- C) MPC used a look-ahead window to anticipate the turn
- D) MPC disabled longitudinal control

ANS: C) MPC used a look-ahead window to anticipate the turn

12. What is one disadvantage of MPC identified in the slides?

- A) It cannot be applied to nonlinear systems
- B) It cannot handle constraints
- C) It never uses optimization
- D) It requires an accurate, computationally efficient model and may be expensive to solve

ANS: D) It requires an accurate, computationally efficient model and may be expensive to solve

13. What is the effect of choosing a larger MPC look-ahead horizon T ?

- A) Potentially better control performance but higher computation overhead

- B) Guaranteed zero computation cost
- C) More myopic behavior and less planning
- D) Elimination of the system model

ANS: A) Potentially better control performance but higher computation overhead

14. What distinguishes dynamics from kinematics?

- A) Dynamics ignores mass and forces
- B) Dynamics considers forces, mass, and inertia that affect motion
- C) Dynamics uses only geometric relationships
- D) Dynamics cannot describe vehicle motion

ANS: B) Dynamics considers forces, mass, and inertia that affect motion

15. What is the state vector of the kinematic bicycle model presented in the slides?

- A) Steering, throttle, and brake
- B) Acceleration, force, and torque
- C) Position x, position y, and heading angle theta
- D) Wheel speed, tire pressure, and yaw torque

ANS: C) Position x, position y, and heading angle theta

16. Which pair represents the control inputs of the presented kinematic bicycle model?

- A) Mass and inertia
- B) Position and heading
- C) Longitudinal and lateral forces
- D) Steering angle and vehicle speed

ANS: D) Steering angle and vehicle speed

17. For the kinematic bicycle model, how is the turning radius R related to wheelbase L and steering angle delta?

- A) $R = L / \tan(\delta)$
- B) $R = L \tan(\delta)$
- C) $R = \delta / L$
- D) $R = vL$

ANS: A) $R = L / \tan(\delta)$

18. What happens to the turning radius as the steering angle approaches zero?

- A) It approaches zero
- B) It approaches infinity, corresponding to straight-line motion
- C) It becomes equal to vehicle speed
- D) It becomes negative regardless of direction

ANS: B) It approaches infinity, corresponding to straight-line motion

19. In the clockwise racetrack convention used in the slides, what does a positive cross-track error indicate?

- A) The car is exactly on the desired trajectory

- B) The car is inside the track region
- C) The car is outside the track region
- D) The car has stopped

ANS: C) The car is outside the track region

20. How does Twiddle tune PID gains?

- A) It requires an exact nonlinear vehicle model
- B) It changes all gains randomly without evaluation
- C) It applies pole placement to a linear model
- D) It systematically adjusts each gain and its step size according to whether the error improves

ANS: D) It systematically adjusts each gain and its step size according to whether the error improves